MIL-L-1225B
6 NOVEMBER 1957
SUPERSEDING
MIL-L-1225A
19 SEPTEMBER 1952

MILITARY SPECIFICATION

LAUNCHER, GRENADE, M7A3

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one type of grenade launcher which, when attached to the Rifle, U. S., Caliber .30 M1, is used to launch grenades and ground signals.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

UU–T–111 — Tape; Paper, Gummed (Sealing and Securing).

PPP–B–591 — Boxes, Fiberboard, Wood-Cleated.

PPP–B–601 — Boxes, Wood, Cleated-Plywood.

PPP–B–621 — Boxes, Wood, Nailed and Lock-Corner.

PPP–B–636 — Boxes, Fiber.

PPP–B–676 — Boxes, Set-Up, Paperboard.

PPP–T–76 — Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant.

MILITARY

JAN–P–100 — Packaging and Packing for Overseas Shipment, General Specification.

MIL–P–116 — Preservation, Methods of.

JAN–C–372 — Cleaner, Rifle Bore.

MIL–P–3420 — Packaging Materials, Volatile Corrosion Inhibitor Treated.

MIL–C–3955 — Cans, Fiber, Spirally Wound.

MIL–W–13855 — Weapons, Small Arms, General Specification for.

FSC 1005

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:01Z
Check the source to verify that this is the current version before use.

MIL–P–14232 — Packaging of Parts, Spare Parts, Equipment and Tools for Weapons and Fire Control, General Specification for.

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 —Marking for Shipment and Storage.

DRAWING

ORDNANCE CORPS

F7266167 — Launcher, Grenade, M7A3.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE

Consolidated Freight Classification Rules.

(Application for copies should be addressed to Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Production samples. On each contract, samples of grenade launchers and parts representing initial production shall be taken by the Government inspector and shall be forwarded prepaid by the contractor to a testing agency designated by the procuring activity. Samples shall be representative of the materiel which has been inspected and provisionally accepted by the Government inspector. (See 6.1.)

3.2 Materials and construction. Grenade launchers and parts shall conform to the materials, dimensions, tolerance limits, physical properties, degree of surface roughness, and final protective finish specified on the drawings listed on Drawing F7266167, and the materials and construction shall be in accordance with Specification MIL–W–13855.

3.3 Design. Grenade launchers and parts shall conform to the design specified on the drawings listed on Drawing F7266167 and to this specification, and shall be in accordance with the applicable design provisions of Specification MIL–W–13855.

3.3.1 *Grenade retention.* The load required to release a launcher from a simulated grenade sleeve shall be within the limits specified on the applicable drawing.

3.3.2 *Functioning.* Grenade launchers shall operate without malfunctions or breakage when subjected to the firing tests prescribed in 4.6.5

3.4 Marking. Grenade launchers and parts shall be marked in accordance with the applicable drawings and the marking provisions of Specification MIL–W–13855.

3.5 Workmanship. Workmanship shall be in acccordance with Specification MIL–W–13855.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions. The quality assurance provisions contained in this section are applicable to inspection and tests performed by or for the Government. Unless otherwise specified herein, inspection shall be performed by the Government inspector, and tests shall be performed by the contractor under the surveillance of the Government inspector. Whenever the term "inspector" is used herein, it denotes Government inspector.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:01Z
Check the source to verify that this is the current version before use.

4.2 **Production samples.** Inspection and testing of production samples (see 3.1) shall be performed at the designated testing agency. Production samples shall be subjected to the inspection and tests specified herein and such other inspection and tests as are necessary to determine compliance with the requirements of the contract. Acceptance of the initial lot shall be deferred pending approval of the initial production samples.

4.3 **Pilot pack.** Inspection and testing of the pilot pack (see 5.1) shall be performed at the designated testing agency. The pilot pack shall be subjected to the inspection and tests specified herein and such other inspection and tests as are necessary to determine compliance with the preservation, packaging, and packing requirements of this specification for levels A and B as applicable. The level A pilot pack shall be subjected to the rough handling test specified in 4.6.6.1. Acceptance of the initial lot shall be deferred pending approval of the pilot pack. The initial production samples may be included in the pilot pack.

4.4 **Sampling.**

4.4.1 *Inspection lot.* The formation, size, and presentation of inspection lots shall be in accordance with Standard MIL–STD–105. The inspector shall insure that inspection lots are as large as practicable, in consideration of quality history, manufacturing conditions, and contractor's delivery schedule, and within the limitations of the lot provisions of Specification MIL–W–13855.

4.4.2 *Inspection samples.*

4.4.2.1 *Grenade launchers.* Sample sizes for inspection of grenade launchers shall be in accordance with Standard MIL–STD–105, using inspection level II, and the inspection sample provisions of Specification MIL–W–13855.

4.4.2.2 *Packaging.* Sample sizes for inspection of packaging shall be in accordance with Standard MIL–STD–105, using inspection level I, and the inspection sample provisions of Specification MIL–W–13855.

4.4.3 *Test samples.*

4.4.3.1 *Grenade launchers.* The number of test samples shall be as specified for each test.

4.4.3.2 *Packaging.* The number of test samples, except for the rough handling test (see 4.6.6.1), shall be in accordance with the inspection level L6 of appendix to Standard MIL–STD–105.

4.5 **Inspection.** Inspection shall be in accordance with the classification of defects specified in 4.5.1, the additional inspection specified in 4.5.2, and the applicable inspection provisions of Specification MIL–W–13855.

4.5.1 *Classification of defects.* Classification of defects and acceptable quality levels (AQL's) for inspection, in accordance with Standard MIL–STD–105, shall be as specified herein and shall be subject to the following provisions:

(a) An individual AQL is specified for each listed defect.

(b) Dimensions are given in the classification of defects for grenade launchers only as a guide in locating defects specified. Refer to latest revision of drawings included in the contract for the acceptable dimensions.

(c) The presence of nicks, burs, cracks, scratches, tool marks, deformations, missing operations, or other signs of poor workmanship on important areas should be carefully considered.

(d) The term "Gage" under "Inspection method" indicates the existence of a Government gage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:01Z
Check the source to verify that this is the current version before use.

(e) Inspection for packaging defects specified in 4.5.1.2 shall apply to each item of the applicable sample of grenade launchers, interior packages, or exterior containers, as applicable.

4.5.1.1 *Launcher, Grenade, M7A3 (dwg. F7266167).* See 4.5.1.2 for inspection of packaging and 4.5.2 for additional inspection.

4.5.1.1.1 *Latch (dwg. C7313282 covering a detail of dwg. F7266167).* Inspect prior to assembly to grenade launcher assembly.

Defect	*AQL*	*Inspection method*	*Dimension*
Major			
None			
Minor			
51. Missing or defective protective coating	1.5	Visual (See 4.6.2)	
52. Improper nibs	1.5	Visual	
53. Distance between arms, open end, over	1.5	Visual	.60 —.06
54. Location of pin hole	1.5	Gage	1.065 + .005
55. Width, inside, rear, under	1.5	Measure	.720 ± .005
56. Diameter of pin hole	2.5	Gage	.168 + .002
101. Workmanship	6.5	Visual	

4.5.1.1.2 *Spring, retainer (dwg. B7266162 covering a detail of dwg. F7266167).* Inspect prior to assembly to grenade launcher assembly.

Defect	*AQL*	*Inspection method*	*Dimension*
Major			
1. Diameter, wire	0.65	Measure	.0410 ± .0005
2. Height	1.0	Measure	.138 + .012
Minor			
51. Missing or defective protective coating	1.5	Visual (see 4.6.2)	
52. Improper contour	1.5	Visual	
53. Length, overall	1.5	Visual	.760 —.020
54. Length, locking legs	1.5	Visual	¼ — 1/32
55. Width, overall	2.5	Measure	.335 ± .010
101. Workmanship	6.5	Visual	

4.5.1.1.3 *Launcher, grenade, assembly (dwg. D7266166 covering a detail of dwg. F7266167).* This classification of defects shall be used only after the application of 4.5.1.1.1, 4.5.1.1.2, and 4.5.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:01Z
Check the source to verify that this is the current version before use.

Defect	AQL	Inspection method	Dimension
Major			
1. Outside diameter, tube, under	0.65	Gage	.863 — .003
2. Functional assembly	0.65	Gage	
3. Alinement of tube	0.65	Gage	
4. Recoil test	0.65	Gage	
5. Distance, rear of latch to recoil stop	0.65	Gage	3.465 (computed)
6. Tube straightness	0.65	Gage	
7. Missing parts	0.65	Visual	3.551 (computed)
8. Improper staking, latch pin	0.65	Visual—manual	
9. Improper riveting or staking of valve stud	0.65	Visual—manual	
10. Length of stud	0.65	Gage	.346 — .005
11. Improper grenade retention (see 3.3.1)	0.65	Gage	5 to 15 pounds
12. Diameter of barrel hole, over	1.0	Gage	.5175 + .002
Minor			
51. Missing or defective protective coating	1.5	Visual (See 4.6.2)	
52. Improper internal contour, rear of barrel hole	1.5	Visual	.523 + .005 10° ± 2°
53. Improper internal contour, front of barrel hole	1.5	Visual	.57 + .01 10° ± 2°
54. Location and marking of rings	1.5	Visual	
55. Improper chamfer, inside hole, front of tube	1.5	Visual	1/16 ± 1/64 45° ± 1°
56. Improper chamfer, outside front of tube	1.5	Visual	1/16 ± 1/64 45° ± 1°
57. Missing or improper identification	1.5	Visual	
58. Inside diameter of tube	1.5	Gage	.485 + .005
59. Physical requirement, body	1.5	Hardness test (See 4.6.3)	Rockwell C20–35
60. Physical requirement, sight pad rest	1.5	Hardness test (See 4.6.3)	Rockwell C50–60
61. Vertical location and squareness of sight pad rest from hole "c"	1.5	Gage	.471 — .007

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:01Z
Check the source to verify that this is the current version before use.

Defect	AQL	Inspection method	Dimension
62. Vertical location of sight boss holes from sight pad rest	1.5	Gage	.470 — .005
63. Diameter of sight boss holes	2.5	Measure	1.87 + .005
64. Width, overall	2.5	Measure	1.250 — .010
65. Protrusion of latch pin	4.0	Measure	.095 minimum
101. Workmanship	6.5	Visual	

4.5.1.2 *Packaging.*

Defect	AQL	Inspection method and requirement
Major		
1. Illegible marking	1.0	Visual (See 5.5)
2. Incorrect marking	1.0	Visual (See 5.5)
3. Improper level of packaging or packing	1.0	Visual
Minor		
51. Inadequate drying	2.5	Visual (See 5.3.1.2, 5.3.2.1, or 5.3.3, as applicable).
52. Improper preservative application and drainage	2.5	Visual (See 5.3.1.3, 5.3.2.1, or 5.3.3, as applicable).
53. Improper wrapping of launcher	2.5	Visual (See 5.3.1.4, 5.3.2.2, or 5.3.3, as applicable).
54. Improper closure of interior packages	2.5	Visual (See 5.3.1.4 and 5.3.1.5, 5.3.2.2 and 5.3.2.3, or 5.3.3, as applicable).
55. Improper closure and strapping of shipping container	2.5	Visual (See 5.4.1, 5.4.2, or 5.4.3, as applicable).
101. Workmanship	6.5	Visual (See 5.7).

4.5.2 *Additional inspection.* In addition to the classification of defects specified in 4.5.1, samples from each inspection lot of M7A3 grenade launchers shall be tested for hardness requirements specified on the drawings. The following parts shall be inspected, prior to assembly, in accordance with Standard MIL–STD–105 using sample size code letter D regardless of lot size. Lot acceptance shall be based on an AQL of 10.0 percent defective. Acceptance and rejection numbers shall be applied to the individual defect, not to a group of defects. Dimensions are given only as a guide in locating defects as specified in 4.5.1(b).

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:01Z
Check the source to verify that this is the current version before use.

Defect	Inspection method	Dimension
201. Physical requirement, latch	Hardness test	Rockwell C30–35
202. Physical requirement, pin, latch	Hardness test	Rockwell A74 maximum
203. Physical requirement, stud, valve	Hardness test	Rockwell C35–50
204. Physical requirement, plunger	Hardness test	Rockwell C35–45

4.6 Tests. In addition to the applicable test provisions specified in Specification MIL–W–13855 and other tests required by applicable drawings and specifications, the following tests shall apply. Unless otherwise specified herein, samples for tests shall be taken by the inspector from each inspection lot. Test fixtures and equipment shall be in accordance with Government drawings or, in the absence of drawings, shall be approved by the procuring agency.

4.6.1 *Material tests.* Tests required by the applicable drawings and specifications shall be performed on materials used in the manufacture of grenade launchers. The contractor shall furnish the inspector certified copies of chemical analyses and reports of physical tests as may be required for each lot of each material used. The contractor shall also furnish the inspector certification that the materials used in packaging conform to the applicable specifications.

4.6.2 *Protective finish tests.* Protective finish tests shall be made in accordance with applicable specifications and drawings. When protective finish testing is specified, samples from each protective finish or inspection lot shall be subjected to this test. Failure of any sample to meet the requirements of this test shall cause rejection of the lot.

4.6.3 *Hardness test.* When prescribed in the applicable classification of defects, samples of parts shall be tested by the inspector for hardness requirements specified on the drawings.

4.6.4 *Depth of case test.* When depth of case is specified on the drawings, case depth shall be measured by the contractor on at least three samples taken from each batch-furnace heat-treat lot, or one sample from each pot each day from each salt pot used. Samples used for case depth measurement may be scrap components or test bars of equivalent analysis and cross section that are heat-treated with the represented lot. Failure of any case depth sample to meet the drawing requirements shall cause rejection of the represented lot.

4.6.5 *Function firing tests.* Firing tests shall be performed on production samples by the testing agency to insure that the functioning requirements have been met.

4.6.5.1 *For initial production samples.* At least five launchers from the initial production sample shall be fired in the following sequence:

(a) 10 rounds, Grenade, AT, Practice, M11A3, using Cartridge, Rifle, Grenade, Cal .30, M3.

(b) 10 rounds, Grenade, AT, Practice, M11A3, using Cartridge, Rifle, Grenade, Cal .30, M3, and Cartridge, Rifle, Grenade, Cal .30, M7, Auxiliary.

(c) 10 rounds, Cartridge, Ball Cal .30, M2, with launcher attached to rifle.

4.6.5.2 *For production samples (except initial production samples).* At least one launcher from each sample shall be fired in the following sequence:

(a) 2 rounds, Grenade, AT, Practice, M11A3, using Cartridge, Rifle,

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:01Z
Check the source to verify that this is the current version before use.

Grenade, Cal .30, M3, and Cartridge, Rifle, Grenade, Cal .30, M7, Auxiliary.

(b) 10 rounds, Cartridge, Ball, Cal .30, M2, with launcher attached to rifle.

4.6.6 *Packaging tests.*

4.6.6.1 *Rough handling test.* The level A pilot pack shall be subjected to the rough handling test specified in Specification JAN–P–100.

4.6.6.2 *Quick leak test.* Each sample level A unit package shall be subjected to the quick leak test specified in Specification MIL–P–116. Lot acceptance shall be based on an AQL of 6.5 percent defective.

4.6.6.3 *Determination of cleanliness test.* Each sample launcher shall be subjected to the determination of cleanliness test specified in Specification MIL–P–116. Lot acceptance shall be based on an AQL of 6.5 percent defective.

4.7 **Reinspection and retests.** Reinspection and retests shall be in accordance with Specification MIL–W–13855.

5. PREPARATION FOR DELIVERY

5.1 **Pilot pack.** When level A or B is specified, a pilot pack consisting of a complete and packed shipping container shall be forwarded prepaid by the contractor to the testing agency designated by the procuring activity for approval. (See 6.1.)

5.2 **Materials and methods.** The materials and methods used in preparing the grenade launcher for shipment and storage shall be in accordance with Specification MIL–P–116 and as specified herein. Interior packages and shipping containers shall be of uniform size and designed to fit the contents in a manner that shall prevent movement in transit.

5.3 **Preservation and packaging.**

5.3.1 *Level A.*

5.3.1.1 *Cleaning.* Each launcher shall be cleaned in accordance with process C–3. Launchers that have been exposed to burned powder residues shall be scrubbed clean with rifle bore cleaner conforming to Specification JAN–C–372 prior to cleaning by process C–3.

5.3.1.2 *Drying.* After cleaning, all launchers shall be dried in accordance with any applicable procedure listed in Specification MIL–P–116.

5.3.1.3 *Preservative application.* Each launcher shall be preserved with type P–9 oil and P–18 volatile corrosion inhibitor (VCI). Type P–9 oil shall be applied by the dipping procedure. Excess oil shall be thoroughly drained from the launcher prior to wrapping. The P–18 preservative shall conform to type I, grade C of Specification MIL–P–3420 and shall be applied as specified in 5.3.1.4.

5.3.1.4 *Unit packaging.* Each launcher shall be completely wrapped with P–18 preservative and unit packaged in accordance with method IA–13. The fiber container shall conform to type I, grade B, style a, class 1 of Specification MIL–C–3955.

5.3.1.5 *Intermediate packaging.* Unless otherwise specified, 50 launchers, packaged as described in 5.3.1.4, shall be placed in a fiberboard box conforming to style RSC, class 3, W5c of Specification PPP–B–636. Seams and joints shall be taped with 2-inch pressure-sensitive tape conforming to Specification PPP–T–76.

5.3.2 *Level B* (see section 6).

5.3.2.1 *Cleaning, drying, and preservative application.* Cleaning, drying, and preservative application shall be in accordance with the requirements of 5.3.1.1, 5.3.1.2, and 5.3.1.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:01Z
Check the source to verify that this is the current version before use.

5.3.2.2 *Unit packaging.* Each launcher shall be completely wrapped with P–18 preservative and unit packaged in accordance with method I. The P–18 preservative shall consist of aluminum-foil-backed VCI-treated material conforming to type I of Specification MIL–P–3420. The launcher shall be placed in a paperboard box conforming to Specification PPP–B–676. The box shall be sealed with gummed kraft tape conforming to Specification UU–T–111.

5.3.2.3 *Intermediate packaging.* Unless otherwise specified, 50 launchers packaged as described in 5.3.2.2 shall be placed in a fiberboard box conforming to type I, class 1, style RSC of Specification PPP–B–636. Seams and joints shall be taped with 3-inch tape conforming to class 2 of Specification UU–T–111.

5.3.3 *Level C.* Cleaning, drying, preservation, and packaging shall be in accordance with the manufacturer's commercial practice. Protection shall be such as to prevent deterioration or damage during shipment and for 3 months' covered storage at the first receiving installation.

5.4 Packing.

5.4.1 *Level A.* Intermediate packages, packaged as described in 5.3.1.5, shall be packed in a wood-cleated plywood box or nailed wood box conforming to the requirements of Specification PPP–B–601, overseas type, or Specification PPP–B–621, class 2, style 2. The gross weight of a fully packed container shall not exceed 200 pounds. The container shall be strapped in accordance with the appendix of the applicable container specification. Complete packs shall be capable of passing the applicable rough handling test specified in 4.6.6.1.

5.4.2 *Level B.* Intermediate packages, packaged as described in 5.3.2.3, shall be packed in a wood-cleated fiberboard box, wood-cleated plywood box, or a nailed wood box conforming to the requirements of Specifications PPP–B–591, domestic type, PPP–B–601, domestic type, or PPP–B–621, class 1. The gross weight of a fully packed container shall not exceed 200 pounds. The container shall be strapped in accordance with the appendix of the applicable container specification.

5.4.3 *Level C.* Launchers, packaged as specified in 5.3.3, shall be packed in a shipping container conforming to Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation. The launchers shall be protected by the use of either fiberboard separators or partitions designed to fit the contents in a manner that shall prevent movement in transit. The quantity per container shall not exceed 100 launchers. The shipping container shall assure adequate protection against damage during shipment and safe delivery at destination.

5.5 Marking. Interior packages and exterior containers shall be marked in accordance with Standard MIL–STD–129. Marking of exterior containers prepared for level A shall be protected with a coating of transparent waterproof material in accordance with Standard MIL–STD–129.

5.6 Spare parts. Spare parts shall be preserved, packaged, and packed in accordance with Specification MIL–P–14232 and other applicable documents.

5.7 Workmanship. Wraps shall be neat and uniform. Tape shall be uniformly applied to interior boxes and shall completely cover and adhere to joints and seams.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:01Z
Check the source to verify that this is the current version before use.

(b) List of applicable drawings and specifications pertinent to the grenade launcher, showing applicable revision dates.

(c) Number of production samples required for tests. (See 3.1.)

(d) Shipping instructions for production samples and pilot pack. (See 3.1 and 5.1.)

(e) Selection of applicable levels of preservation and packaging, and packing.

(1) Level B preservation and packaging is intended to provide adequate and economical protection during multiple domestic shipments, handling, and indoor storage periods up to 12 months from date of initial packaging. (See 6.2.)

(f) Place of final inspection and acceptance. (See "Notes" in Specification MIL–W–13855.)

(g) Responsibility for furnishing manufacturing, inspection, and test equipment. (See "Notes" in Specification MIL–W–13855.)

(h) Responsibility for furnishing ammunition. (See "Notes" in Specification MIL–W–13855.)

(i) Contractor-Government agreement for quality assurance provisions.

6.2 Storage surveillance. Grenade launchers preserved and packaged in accordance with level B requirements must be inspected to determine condition when not used within a 1-year period. Launchers not used within this period should be represerved or repackaged in accordance with this specification as determined necessary by storage conditions and anticipated storage time.

Notice. This specification, together with specifications and drawings pertaining to it and bearing a "Notice" of similar restrictions, is intended for use only in connection with procurement by the United States Government, and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, nor be used for any other purpose except when specifically authorized by the Chief of Ordnance, Department of the Army.

Custodians:
Army—Ordnance Corps
Navy—Bureau of Ordnance

Other interest:
Navy—MC

Preparing activity:
Army—Ordnance Corps

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:01Z
Check the source to verify that this is the current version before use.